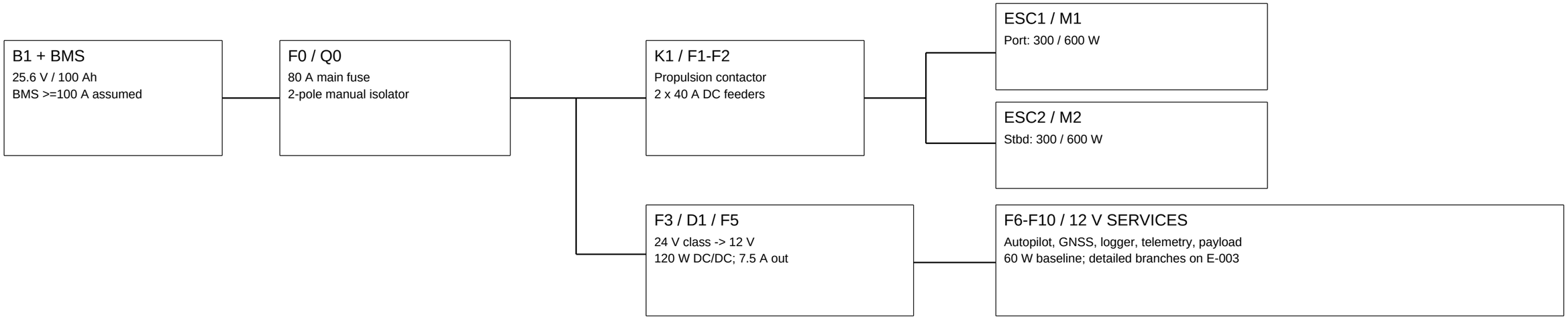
E-001 DESIGN BASIS, SINGLE-LINE ARCHITECTURE AND POWER BUDGET

Independent conceptual study — unbuilt and unvalidated.

1. DESIGN BASIS

Mission: 2 h sheltered-water environmental survey; visual supervision and recovery support.
Nonconductive 2-3 m hull assumed. Displacement, speed, thrust and battery mass are unresolved.
B1: 8S LiFePO4 / 25.6 V nominal / 100 Ah / planning range 22.4-29.2 V.
No charger aboard. Remove/isolate pack for approved shore charging. No AC system in scope.

2. FUNCTIONAL SINGLE-LINE (RETURNS OMITTED ON THIS SHEET)



3. POWER BUDGET - ASSUMED ELECTRICAL INPUTS

LOAD	CRUISE W	PEAK W	BASIS
Propulsion (2 thrusters)	600	1200	2 x 300 / 600 W
12 V services	60	60	D1 output load
D1 conversion loss	6.7	6.7	90% efficiency
Permit/control coils	4	4	Allowance incl. relay drives
TOTAL AT BATTERY	670.7	1270.7	Simultaneous peak

ENERGY ESTIMATE (NOT MEASURED)

Nominal: 25.6 x 100 = 2560 Wh

Planning: 2560 x 0.80 x 0.90 = 1843 Wh

Mission: 670.7 x 2 = 1341 Wh

Endurance: 1843 / 670.7 = 2.75 h

At 900 W propulsion: only 1.90 h

Cable/contact losses not explicitly added.

See calculations/calculations.md.

ENV-USV-01 | CIVILIAN WATER-QUALITY MONITORING

Design basis, single-line architecture and power budget

Prepared for: Pratik Prabhakar Shinde | AI-assisted independent study

DXF generated with Python. Not drafted or validated inside AutoCAD.

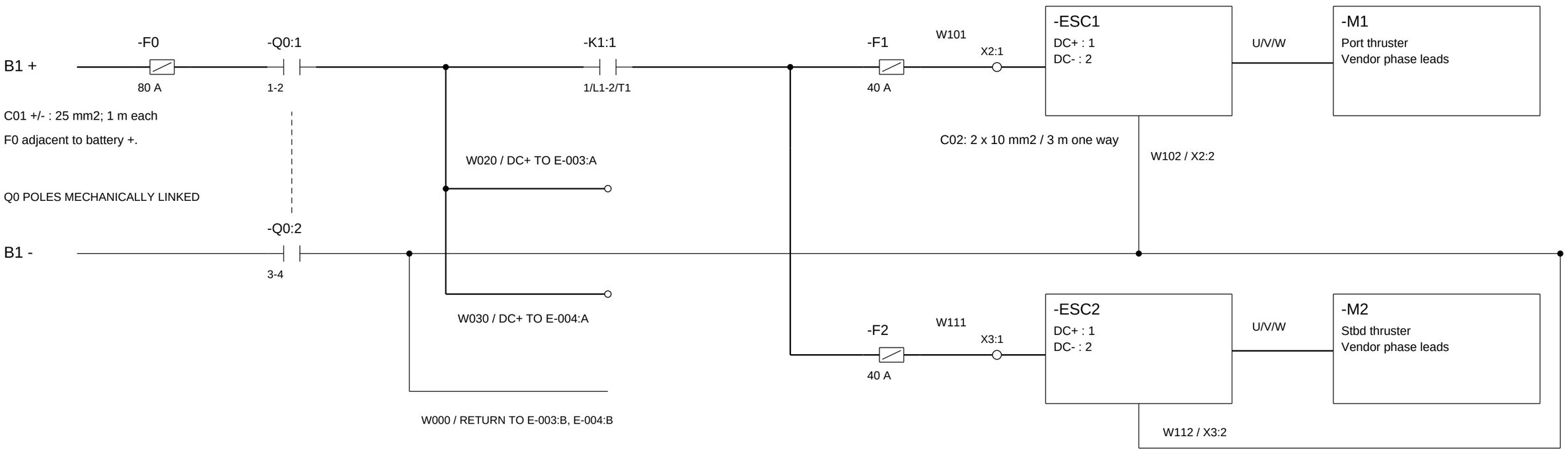
DRAWING: E-001
REV: B / 17 OCT 2026
CHECK: NOT REVIEWED
APPROVAL: NOT ISSUED

A3 / 420 x 297 mm
SCALE: NTS
UNITS: mm (sheet)
PLOT: 1:1 / LANDSCAPE

E-002 MAIN DC DISTRIBUTION AND PROPULSION FEEDERS

Independent conceptual study — unbuilt and unvalidated.

POWER CIRCUITS / SIMPLIFIED DEVICE SYMBOLS / ALL CONTACTS SHOWN DE-ENERGIZED



FEEDER / PROTECTION CALCULATION BASIS

CIRCUIT	I (A)	1.25 I	mm2	FUSE A	DROP V
C01 main	56.73	70.91	25	80	0.102
C02 / C03	26.79	33.48	10	40	0.362 each
C04 D1 in	5.95	7.44	2.5	10	0.214 at 120 W

I = P / 22.4 V; Rloop = 2 L rho / A

rho = 0.0225 ohm mm2/m (assumed)

Main + motor feeder: 0.464 V / 2.07%

Excludes contact/fuse/switch losses.

Wire numbers identify conductors, not vendor pins.

No dot at wire crossing = no connection.

HOLD: cable ampacity / battery fault current / fuse interrupt rating and selectivity / ESC inrush, UV and regeneration.

Switchgear voltage rating >=60 V DC is a target; actual DC breaking capability and fault withstand must be checked.

ENV-USV-01 | CIVILIAN WATER-QUALITY MONITORING

Main DC distribution and propulsion feeders

Prepared for: Pratik Prabhakar Shinde | AI-assisted independent study

DXF generated with Python. Not drafted or validated inside AutoCAD.

DRAWING: E-002

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CHECK: NOT REVIEWED

APPROVAL: NOT ISSUED

A3 / 420 x 297 mm

SCALE: NTS

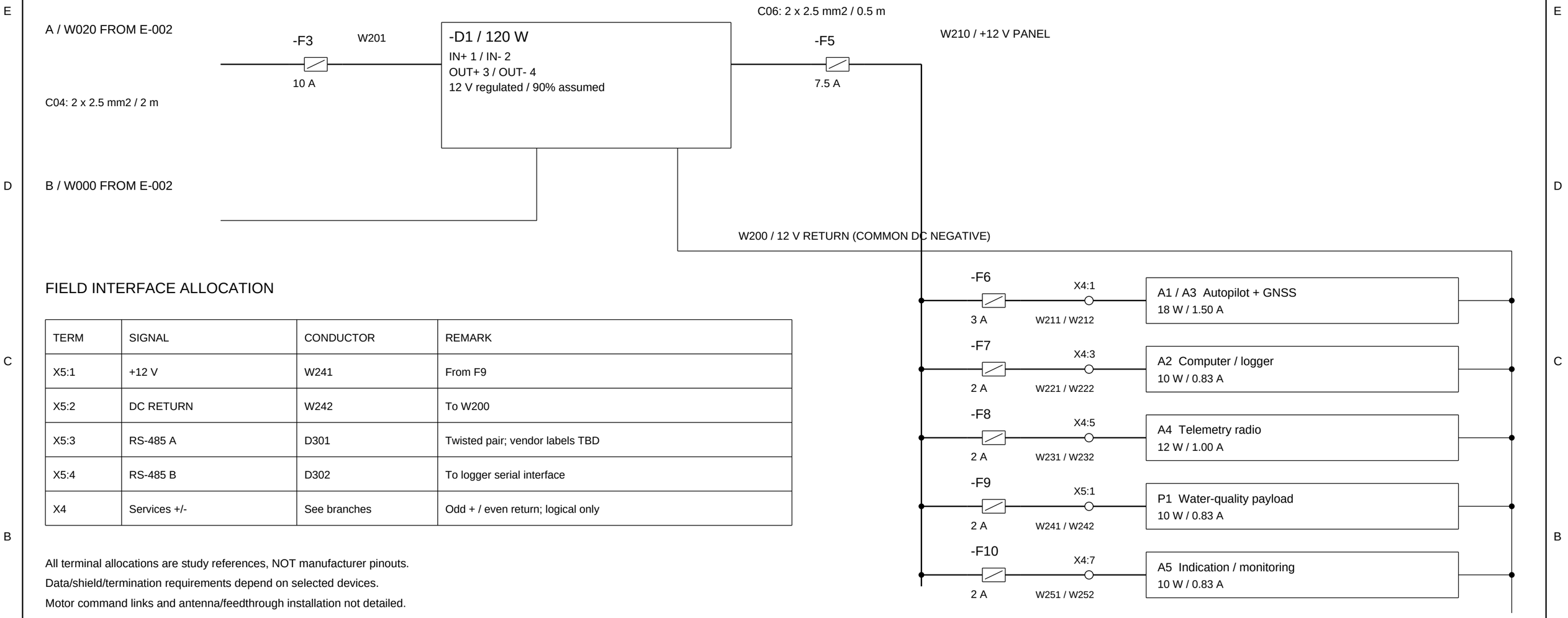
UNITS: mm (sheet)

PLOT: 1:1 / LANDSCAPE

E-003 12 V SERVICES, FIELD TERMINALS AND SENSOR INTERFACES

Independent conceptual study — unbuilt and unvalidated.

24 V CLASS TO 12 V SUPPLY / NON-ISOLATED CONVERTER / DEDICATED COPPER RETURNS



FIELD INTERFACE ALLOCATION

TERM	SIGNAL	CONDUCTOR	REMARK
X5:1	+12 V	W241	From F9
X5:2	DC RETURN	W242	To W200
X5:3	RS-485 A	D301	Twisted pair; vendor labels TBD
X5:4	RS-485 B	D302	To logger serial interface
X4	Services +/-	See branches	Odd + / even return; logical only

All terminal allocations are study references, NOT manufacturer pinouts.
Data/shield/termination requirements depend on selected devices.
Motor command links and antenna/feedthrough installation not detailed.

BRANCH WIRING: <=2 m one way; + and return each 0.75 mm2. Baseline 12 V load = 5.0 A.
Worst branch drop 0.180 V; plus C06 0.045 V = 0.225 V / 1.88%. Contact losses extra.
F5 limits usable panel load below D1 full 120 W rating. Recheck protection before adding payloads.

ENV-USV-01 | CIVILIAN WATER-QUALITY MONITORING

12 V services, field terminals and sensor interfaces

Prepared for: Pratik Prabhakar Shinde | AI-assisted independent study

DXF generated with Python. Not drafted or validated inside AutoCAD.

DRAWING: E-003
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CHECK: NOT REVIEWED
APPROVAL: NOT ISSUED

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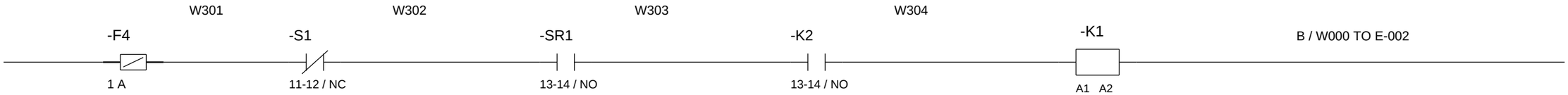
PLOT: 1:1 / LANDSCAPE

E-004 PROPULSION PERMIT CONCEPT AND COMMISSIONING CHECKS

Independent conceptual study — unbuilt and unvalidated.

HARDWARE PERMIT PATH - FUNCTIONAL CONTACT DIAGRAM, NOT A RELEASED SAFETY CIRCUIT

A / W030 FROM E-002



C05: 0.75 mm2; 2 m equivalent loop assumption; suppression and relay supply wiring are not detailed.

-SR1 / MANUAL PERMIT

Functional module; not selected.
Requires deliberate reset/arm action.
No auto re-arm after power/stop reset.
Reset and feedback terminals TBD.

-K2 / WATCHDOG PERMIT

NO contact only while heartbeat valid.
>1 s timeout -> neutral, then open K1.
HW/SW implementation not provided.
Failure-high behaviour unresolved.

-K1 / PROPULSION CONTACTOR

Main contact on E-002; hotel stays live.
Auxiliary feedback needed for fault detection.
Coil suppression / drop-out time TBD.
Welded contact can defeat isolation.

PROPOSED TESTS - ALL NOT RUN / RECORD INSTRUMENTS, CONDITIONS, VALUES, PASS/FAIL AND EVIDENCE

TEST	METHOD	PROPOSED ACCEPTANCE / HOLD POINT
T01-03	Pre-energization review	Datasheets, fault current, polarity, torque, return continuity and BMS limits approved.
T04	Current-limited supply	12 V panel remains 11.4-12.6 V at 60 W; no reset over intended input range.
T05	Guarded load emulation	Cable-only propulsion path drop <=3%; ESC input above UV limit; thermal limits met.
T06-07	Stop / heartbeat loss	Proposed K1 power removal <=0.5 s on local stop; >1 s timeout response; manual re-arm.
T08-09	Ingress / wet motor setup	Approved facility/procedure; record transients; fault/feedback response reviewed.
T10	Supervised water trial	2 h within pack/SOC/thermal limits; log V/I/Wh and conditions; recovery support.

No deliberate pack short circuit. No dry-running water-cooled thrusters. Power removal does not prove stopped thrust or vessel motion.

ENV-USV-01 | CIVILIAN WATER-QUALITY MONITORING

Propulsion permit concept and commissioning checks

Prepared for: Pratik Prabhakar Shinde | AI-assisted independent study

DXF generated with Python. Not drafted or validated inside AutoCAD.

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E-005 CABLE SCHEDULE, PRELIMINARY MATERIALS AND DRAWING NOTES

Independent conceptual study — unbuilt and unvalidated.

CABLE TAKE-OFF - CONDUCTOR-METRES, INCLUDING 20% ROUTING ALLOWANCE

ID	FROM / TO	L ONE-WAY	CORE mm2	CORES	PROV. m	PROTECTION / HOLD
C01	B1 / F0 / Q0 / DC bars	1.0 m	25	2	2.4	F0 80 A / assumed Iz 100 A
C02	K1 / F1 / ESC1 + return	3.0 m	10	2	7.2	F1 40 A / assumed Iz 50 A
C03	K1 / F2 / ESC2 + return	3.0 m	10	2	7.2	F2 40 A / assumed Iz 50 A
C04	F3 / D1 input + return	2.0 m	2.5	2	4.8	F3 10 A / assumed Iz 20 A
C05	F4 / stop-permit / K1 coil	2.0 m eq.	0.75	2 eq.	4.8	F4 1 A / assumed Iz 5 A
C06	D1 / F5 / 12 V panel	0.5 m	2.5	2	1.2	F5 7.5 A / assumed Iz 20 A
12V-BR	F6-F10 / service loads	2 m each	0.75	2 x 5	24.0	F6 3 A; F7-F10 2 A

PRELIMINARY FUNCTIONAL BOM (NO PART NUMBERS OR COST CLAIM)

TAG	QTY	FUNCTIONAL SPECIFICATION	RELEASE HOLD
B1 / F0 / Q0	1 set	25.6 V 100 Ah; 80 A fuse; >=100 A isolator	Fault current, BMS, DC break, mass
K1 / ESC / M	1/2/2	>=100 A contactor; 600 W peak per propulsion branch	ESC UV <=21 V; inrush / regen / cooling
D1 / F3-F10	1 set	12 V 120 W DC/DC and branch fuse/holder set	Actual voltage range, protection coordination
S1 / SR1 / K2	1 each	NC local stop; manual permit; watchdog permit	Exact reset / feedback and fault behaviour
A1-A5 / P1	1 set	Autopilot, GNSS, logger, radio, monitoring, sensors	Supply compatibility, data protocols and EMC
ENC / cabling	1 lot	Sealed enclosure, glands, connectors, copper, terminals	IP67 assembly target; heat / corrosion review

GENERAL DRAWING / DESIGN NOTES

1. NTS schematic sheets. Model-space coordinates are paper mm. Open DXF in AutoCAD; plot A3 landscape 1:1.
2. Simplified conventional symbols only. Tags/terminal allocations are illustrative, not vendor connection instructions.
3. All ampacities are assumptions. Validate installation derating, connector ratings and fuse curves before design release.
4. Single battery, BMS and converter remain common failure points. No redundancy or standards compliance is claimed.
5. Rev B replaces rounded infographic blocks with CAD entities, tagged circuits and terminal references; no hardware change.
6. Sources: Victron Wiring Unlimited, DC wiring; Blue Sea Systems Circuit Protection. Full links in docs/engineering_brief.md.
7. No drawings created in AutoCAD, hardware measurements, dynamic simulation or employer involvement are claimed.

ENV-USV-01 | CIVILIAN WATER-QUALITY MONITORING

Cable schedule, preliminary materials and drawing notes

Prepared for: Pratik Prabhakar Shinde | AI-assisted independent study

DXF generated with Python. Not drafted or validated inside AutoCAD.

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